GATE Data Science – Mock Practice (06.09.2026)

NPTEL · 65 questions · correct answers highlighted in green

Question 1 / 65 1 mark

In an undirected connected planar graph G, there are eight vertices and five faces. The number of edges in G is _____.

Answer: **11**

Question 2 / 65 1 mark

Consider the following python function:

```
def fun(n):
    for i in range(1, n + 1):
        j = 1
        while j < n:
            print(i, j)
            j += i
```

Time complexity of fun in terms of Θ notation is

A. $\Theta(n^n)$

B. $\Theta(n^2)$

C. $\Theta(n \log n)$ ✓ **Correct**

D. $\Theta(n^2 \log n)$

Question 3 / 65 1 mark

Suppose a stack implementation supports an instruction REVERSE, which reverses the order of elements on the stack, in addition to the PUSH and POP

instructions. Which one of the following statements is TRUE (with respect to this modified stack)?

A. A queue cannot be implemented using this stack.

B. A queue can be implemented where ENQUEUE takes a single instruction and DEQUEUE takes a sequence of two instructions.

C. A queue can be implemented where ENQUEUE takes a sequence of three instructions and DEQUEUE takes a single instruction. ✓ **Correct**

D. A queue can be implemented where both ENQUEUE and DEQUEUE take a single instruction each.

Question 4 / 65 1 mark

Select the correct statement(s).

A. The complexity of Selection sort remains the same irrespective of the sequence of elements. ✓ **Correct**

B. The complexity of Insertion sort remains the same irrespective of the sequence of elements.

C. The complexity of Merge sort remains the same irrespective of the sequence of elements. ✓ **Correct**

D. The complexity of Quicksort remains the same irrespective of the choice of pivot.

Question 5 / 65 1 mark

What is the output of the following snippet of code?

```
def foo(n):
   if n % 2 == 0:
        return n // 2
     return 3 * n + 1
count = 0
x = 10
while x != 1:
        x = foo(x)
        count += 1
print(count)
```

Answer: **6**

Question 6 / 65 1 mark

Consider the following program written in pseudo-code. Assume that x and y are integers:

```
Count(x, y) {
   if (y != 1) {
     if (x != 1) {
       print(“*”);
       Count(x/2, y);
} else {
     y = y – 1;
     Count(1024, y);
     }
  }
}
```

The number of times that the print statement is executed by the call Count(1024, 1024) is _____.

Answer: **10230**

Question 7 / 65 1 mark

Let A be a priority queue for maintaining a set of elements. Suppose A is implemented using a max-heap data structure. The operation EXTRACT-MAX(A) extracts and deletes the maximum element from A. The operation INSERT(A, key) inserts a new element key in A. The properties of a max-heap are preserved at the end of each of these operations. When A contains n elements, which one of the following statements about the worst-case running time of these two operations is TRUE?

A. Both EXTRACT-MAX(A) and INSERT(A, key) run in $O(1)$.

B. Both EXTRACT-MAX(A) and INSERT(A, key) run in $O(\log n)$. ✓ **Correct**

C. EXTRACT-MAX(A) runs in $O(1)$ whereas INSERT(A, key) runs in $O(n)$.

D. EXTRACT-MAX(A) runs in $O(1)$ whereas INSERT(A, key) runs in $O(\log n)$.

Question 8 / 65 1 mark

Consider the Python program fragment below, which is meant to divide x by y using repeated subtractions. The variables x, y, q, and r are all unsigned int.

```
while r >= y:
r = r - y
q = q + 1
```

Which of the following conditions on the variables x, y, q, and r before the execution of the fragment will ensure that the loop terminates in a state satisfying

the condition x == (y * q + r)?

A. q == r and r == 0

B. x > 0 and r == x and y > 0 ✓ **Correct**

C. q == 0 and r == x and y > 0

D. q == 0 and y > 0

Question 9 / 65 1 mark

What is the worst-case time complexity of inserting n elements into an empty linked list, maintaining sorted order?

A. $\Theta(n)$

B. $\Theta(n \log n)$

C. $\Theta(n^2)$ ✓ **Correct**

D. $\Theta(1)$

Question 10 / 65 1 mark

What is the number of arithmetic operations required to implement binary search on a sorted array?

A. $\Theta(n)$

B. $\Theta(\log n)$ ✓ **Correct**

C. $\Theta(1)$

D. $\Theta(n^2)$

Question 11 / 65 1 mark

Consider two relations s(A, B, C) and r(P, Q, C) such that relation s has 30

rows and relation r has 12 rows. What is the maximum number of rows that

are possible in r ⋈ s?

Answer: **360**

Question 12 / 65 1 mark

Consider the following table which has three attributes: X, Y, and Z, where X is the primary key and Z is the foreign key referencing X.

X	Y	Z
1	1	5
2	2	5
3	1	5
4	2	4
5	2	4
6	1	1
7	4	1
8	4	7

Which of the following sets of tuples are additionally deleted when the tuple (1, 1, 5) is deleted and ON DELETE CASCADE construct is applied over the table?

A. (2, 2, 5) and (3, 1, 5)

B. (6, 1, 1) and (7, 4, 1)

C. (6, 1, 1), (7, 4, 1), and (8, 4, 7) ✓ **Correct**

D. (6, 1, 1), (7, 4, 1), and (2, 2, 5)

Question 13 / 65 1 mark

Consider the table below and predict the output of the query that follows.

class	house_name	no_of_student
Class 9	Aravali	50
Class 10	Nilgiri	40
Class 11	Shiwalik	60
Class 12	Nilgiri	55

```
SELECT COUNT(house_name)
FROM ((SELECT class, house_name
       FROM house_distribution) as X
       NATURAL JOIN (SELECT house_name, no_of_student
                     FROM house_distribution) as Y);
```

Answer: **6**

Question 14 / 65 1 mark

Consider the following tables T1 and T2.

Table T1

P	Q
2	2
3	8
7	3
5	8
6	9
8	5
9	8

Table T2

R	S
2	2
8	3
3	2
9	7
5	7
7	2

The following integrity constraints are defined:

- In T1, P is the primary key and Q is a foreign key referencing R in T2, with ON DELETE CASCADE and ON UPDATE CASCADE.
- In T2, R is the primary key and S is a foreign key referencing P in T1, with ON DELETE SET NULL and ON UPDATE CASCADE.

If the tuple with P = 8 is deleted from table T1, determine the number of additional tuples (excluding the explicitly deleted tuple) that will be deleted from T1 due to the above referential integrity constraints.

Answer: **0**

Question 15 / 65 1 mark

Consider the relational scheme:

R(E,F,G,H,I,J,K,L,M,N)

with the following set of functional dependencies:

EF→G

F→I,J

EH→K,L

K→M

L→N

Which of the following attribute sets is a candidate key for the relation R?

A. {E, F}

B. {E, F, H} ✓ **Correct**

C. {E, F, H, K, L}

D. {E}

Question 16 / 65 1 mark

Consider the following log sequence for two transactions T1 and T2 operating on a bank database.
The initial balance of the bank account B is ₹12,000.

- Transaction T1 transfers ₹2,000 from account B to a mortgage account M.
- Transaction T2 applies a 5% interest to account B.

The log records are shown below:

1. `T1: start`
2. `T1: B old = 12000, new = 10000`
3. `T1: M old = 0, new = 2000`
4. `T1: commit`
5. `T2: start`
6. `T2: B old = 10000, new = 10500`
7. `T2: commit`

Assume that the database system crashes immediately before log record 7 is written.

When the system restarts and the recovery procedure is invoked, which one of the following statements is correct?

A. We must redo log record 6 to set B to 10500

B. We must undo log record 6 to set B to 10000 and then redo log records 2 and 3 ✓ **Correct**

C. We need not redo log records 2 and 3 because transaction T1 has committed

D. We can apply redo and undo operations in arbitrary order because they are idempotent

Question 17 / 65 1 mark

A company maintains records of sales made by its salespersons and pays them commission based on each individual's total sales made in a year. This data is maintained in a table with following schema:

salesinfo = (salespersonid, totalsales, commission)

In a certain year, due to better business results, the company decides to further reward its salespersons by enhancing the commission paid to them as per the following formula:

If commission ≤ 50000, enhance it by 2%
If 50000 < commission ≤ 100000, enhance it by 4%
If commission > 100000, enhance it by 6%

The IT staff has written three different SQL scripts to calculate enhancement for each slab, each of these scripts is to run as a separate transaction as follows:

T1:

```
UPDATE salesinfo
SET commission = commission * 1.02
WHERE commission <= 50000;
```

T2:

```
UPDATE salesinfo
SET commission = commission * 1.04
WHERE commission > 50000 and commission is <= 100000;
```

T3:

```
UPDATE salesinfo
SET commission = commission * 1.06
WHERE commission > 100000;
```

Which of the following options of running these transactions will update the commission of all salespersons correctly

A. Execute T1 followed by T2 followed by T3

B. Execute T2, followed by T3; T1 running concurrently throughout

C. Execute T3 followed by T2; T1 running concurrently throughout

D. Execute T3 followed by T2 followed by T1 ✓ **Correct**

Question 18 / 65 1 mark

In a schema with attributes A, B, C, D and E following set of functional dependencies are given

- A → B
- A → C
- CD → E
- B → D
- E → A

Which of the following functional dependencies is NOT implied by the above set?

A. CD → AC

B. BD → CD ✓ **Correct**

C. BC → CD

D. AC → BC

Question 19 / 65 1 mark

Consider the set of relations shown below and the SQL query that follows.

Students: (Roll_number, Name, Date_of_birth)
Courses: (Course_number, Course_name, Instructor)
Grades: (Roll_number, Course_number, Grade)

```
SELECT DISTINCT Name
FROM Students, Courses, Grades
WHERE Students.Roll_number=Grades.Roll_number
AND Courses.Instructor = 'Korth'
AND Courses.Course_number = Grades.Course_number
AND Grades.Grade = 'A'
```

Which of the following sets is computed by the above query?

A. Names of students who have got an A grade in all courses taught by Korth

B. Names of students who have got an A grade in all courses

C. Names of students who have got an A grade in at least one of the courses taught by Korth ✓ **Correct**

D. None of the above

Question 20 / 65 1 mark

A clustering index is defined on the fields which are of type:

A. non-key and ordering ✓ **Correct**

B. non-key and non-ordering

C. key and ordering

D. key and non-ordering

Question 21 / 65 1 mark

Despite his initial hesitation, Rehman’s ____ to contribute to the success of the project never wavered.

A. ambivalence

B. satisfaction

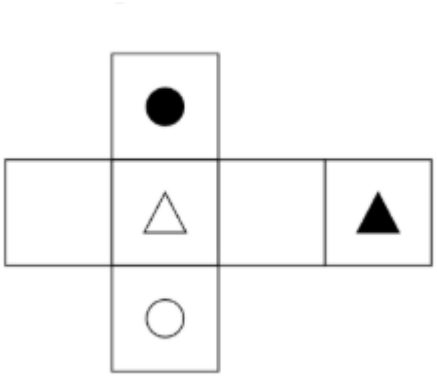
C. resolve ✓ **Correct**

D. revolve

Question 22 / 65 1 mark

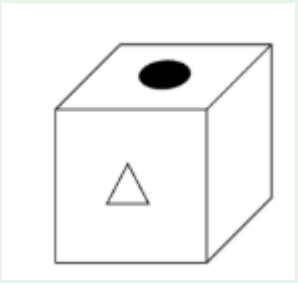
The paper as shown in the figure is folded to make a cube where each square corresponds to a particular face of the cube. Which one of the following options correctly represents the cube?

Note: The figures shown are representative.

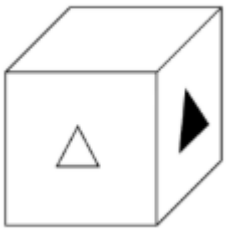


The options are given below:

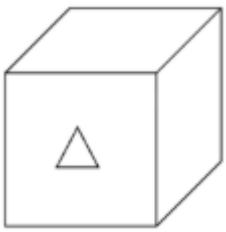
A. ✓ **Correct**



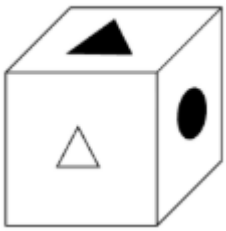
B.



C.



D.



Question 23 / 65 1 mark

If IMAGE and FIELD are coded as FHBNJ and EMFJG respectively, which one among the following options is the most appropriate for BEACH?

A. CEADP

B. IDBFC ✓ **Correct**

C. JGIBC

D. IBCEC

Question 24 / 65 1 mark

Which one of the following options is correct for the given data in the table?

Iteration (i)	Input (I)	Output (X)	Output (Y)
0	20	20	20
1	-4	16	-80
2	10	26	-800
3	15	41	-12000

A. X(i) = X(i − 1) + I(i); Y (i) = Y (i − 1)I(i); i > 0 ✓ **Correct**

B. X(i) = X(i − 1)I(i); Y (i) = Y (i − 1) + I(i); i > 0

C. X(i) = X(i − 1)I(i); Y (i) = Y (i − 1)I(i); i > 0

D. X(i) = X(i − 1) + I(i); Y (i) = Y (i − 1)I(i − 1); i > 0

Question 25 / 65 1 mark

Ravi had ____ younger brother who taught at ____ university. He was widely regarded as ____ honorable man.

Select the option with the correct sequence of articles to fill in the blanks.

A. a; a; an ✓ **Correct**

B. the; an; a

C. a; an; a

D. an; an; a

Question 26 / 65 1 mark

The average marks obtained by a class in an examination were calculated as 30.8. However, while checking the marks entered, the teacher found that the marks of one student were entered incorrectly as 24 instead of 42. After correcting the marks, the average becomes 31.4.

How many students does the class have?

A. 25

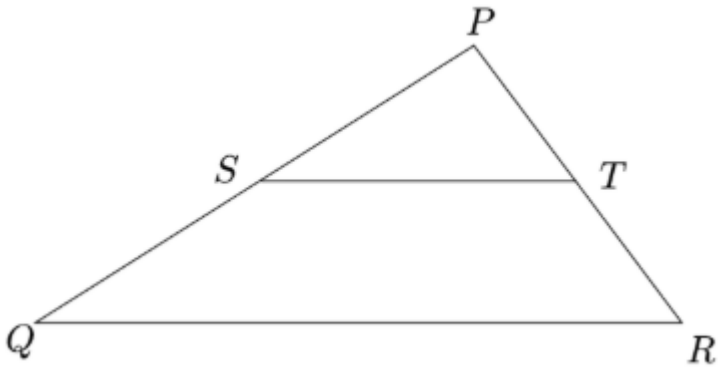
B. 28

C. 30 ✓ **Correct**

D. 32

Question 27 / 65 1 mark

In the diagram, the lines QR and ST are parallel to each other. The shortest distance between these two lines is half the shortest distance between the point P and line QR.



What is the ratio of the area of the triangle PST to the area of the trapezium SQRT?

A. 1/3 ✓ **Correct**

B. 1/4

C. 2/5

D. 1/2

Question 28 / 65 1 mark

The number of coins of Rs1, Rs 5, and Rs10 denominations that a person has are in the ratio 5:3:13. Of the total amount, the percentage of money in Rs 5 coins is

A. 21%

B. $14\ 2/_{7}$

C. 10% ✓ **Correct**

D. 30%

Question 29 / 65 1 mark

A rectangular paper of 20 cm × 8 cm is folded 3 times. Each fold is made along the line of symmetry, which is perpendicular to its long edge. The perimeter of the final folded sheet (in cm) is

A. 21

B. 20

C. 24

D. 18 ✓ **Correct**

Question 30 / 65 1 mark

We reached the station late and _____ missed the train.

A. near

B. nearly ✓ **Correct**

C. utterly

D. mostly

Question 31 / 65 1 mark

Let f and g be two real valued functions such that g(x) = f(x) − 2. Select all true options.

A. If f is odd, then g is odd.

B. If f is even, then g is even. ✓ **Correct**

C. If g is odd, then f is odd.

D. If g is even, then f is even. ✓ **Correct**

Question 32 / 65 1 mark

Evaluate the following limit:

$$\lim_{x \to 2} \frac{x^6 - 24x - 16}{x^3 + 2x - 12}$$

Answer: **12**

Question 33 / 65 1 mark

Find the global minima of the function f(x) = $2x^3-9x^2+12x+6$ in the interval (1, 3). You need to enter the value of x* and not the value of f(x*).

Answer: **2**

Question 34 / 65 1 mark

Define a function f as follows:

$$f(x)=\begin{cases} kx^2+l, & \text{if } x \leqslant 1 \\ lx^2+kx+m, & \text{if } x>1 \end{cases}$$

If f is continuous and differentible at x = 1, find the value of k − 2*l* + m.

Answer: **0**

Question 35 / 65 1 mark

Consider a function f defined as:

$$f(x)=\begin{cases} 3ax+b & x<1, \\ 11 & x=1, \\ 5ax-2b & x>1 \end{cases}$$

If f is continuous at x = 1, then find the value of a + b.

Answer: **5**

Question 36 / 65 1 mark

Select all options that give the correct Taylor series expansion of the function on the left around the point 0.

A. $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$ ✓ **Correct**

B. $\sin x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$

C. $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$ ✓ **Correct**

D. $\cos x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$

Question 37 / 65 1 mark

The height of a right circular cone of maximum volume that can be enclosed within a hollow sphere of radius R is:

A. R

B. 5/4 R

C. 4/3 R ✓ **Correct**

D. 3/2 R

Question 38 / 65 1 mark

Evaluate:

$$\lim_{x \to -3} \frac{\sqrt{2x+22} - 4}{x+3}$$

Answer: **0.25**

Question 39 / 65 1 mark

Find the angle between the tangents to the curve $y = \frac{\pi \sin x}{x}$ at $x = \pi$ and $x = -\pi$.

A. 30°

B. 45°

C. 60°

D. 90° ✓ **Correct**

Question 40 / 65 1 mark

Find the domain and range of $f(x) = \sqrt{5 - 4x - x^2}$.

A. Domain is [0, ∞)

B. Domain is $(-\infty, -5] \cup [1, \infty)$

C. Domain is [−5, 1] ✓ **Correct**

D. Range is [0, ∞)

E. Range is [0, 3] ✓ **Correct**

Question 41 / 65 1 mark

A professor is creating an exam of eleven questions from a test bank of 15 questions. In how many ways can he select and arrange the questions?

A. 15!

B. 15!/11!4!

C. 15!/11!

D. 15!/4! ✓ **Correct**

Question 42 / 65 1 mark

Out of the students in a class, 60% do well in mathematics, 70% do well in physics and 50% do well in both. Find the probability that a randomly selected student doesn't do well either in mathematics or physics.

Answer: **0.2**

Question 43 / 65 1 mark

Students from 3 colleges X, Y , and Z have participated in a competition, where 30% of the participants are from college X, 20% from college Y , and 50% from college Z. It is known that 50%, 40% and 60% of the participants from colleges X, Y , and Z respectively are girls. If a girl is randomly selected, what is the probability that she is from college X? (Enter the answer correct upto three decimal places.)

Answer: **0.283**

Question 44 / 65 1 mark

If a fair coin is tossed an even number 2n times, then what is the probability of obtaining more heads than tails?

A. $^{2n}C_n \left(\frac{1}{2}\right)^{2n}$

B. $\frac{1}{2}\left\{1 - {}^{2n}C_n \left(\frac{1}{2}\right)^{2n}\right\}$ ✓ **Correct**

C. $1 - {}^{2n}C_n \left(\frac{1}{2}\right)^{n}$

D. $^{2n}C_n \left(\frac{1}{2}\right)^{n}$

Question 45 / 65 1 mark

Suppose the number of visitors arriving at a zoo can be modeled to be Poisson distributed. On an average 20 visitors arrive per hour. Let X be the number of visitors arriving from 2pm to 4pm. Then the probability that at least 35 visitors will arrive in the given duration is

A. $\sum_{k=35}^{\infty} \frac{e^{-20}(20)^k}{k!}$

B. $1 - \sum_{k=0}^{34} \frac{e^{-20}(20)^k}{k!}$

C. $\sum_{k=35}^{\infty} \frac{e^{-40}(40)^k}{k!}$ ✓ **Correct**

D. $1 - \sum_{k=0}^{34} \frac{e^{-40}(40)^k}{k!}$ ✓ **Correct**

Question 46 / 65 1 mark

The average life of a coffee machine is 5 years, with a standard deviation of 1 year. Assuming that the lives of these machines follow approximately a normal distribution, find the probability that the mean life of a random sample of 16 such machines falls between 4.6 and 5.4 years. Enter your answer correct to two decimals.

Answer: **0.89**

Question 47 / 65 1 mark

The average diameter of pipes manufactured by a company is reported to be

25mm, which you suspect to be too low. You observed a random sample of diameters of 10 pipes (in mm) as 24.58, 24.67, 24.99, 25.22, 25.32, 25.25, 24.96, 24.87, 25.52, 24.85.
Based on this information, what conclusion can you reach about your suspicion at significance level of 5%?

Use $F_{t_9}^{-1}(0.95) = 1.8331129$

A. Accept H_0 ✓ **Correct**

B. Reject H_0

Question 48 / 65 1 mark

If X and Y are independent Poisson variates such that

P (X = 1) = P (X = 2) and P(Y = 2) = P (Y = 3). Find the variance of $X - 2Y$.

Answer: **14**

Question 49 / 65 1 mark

Let the sample mean of a set of 20 data points be 30. If it is discovered that a data point having value 25 was incorrectly entered as 15, what should be the revised value of the sample mean? Enter the answer correct to one decimal place.

Answer: **30.5**

Question 50 / 65 1 mark

Let X, Y ~ Uniform{1, 2} be i.i.d discrete random variables. Define U = X +Y and V = X. Find the correlation coefficient between U and V . Enter your answer correct to two decimal places.

Answer: **0.71**

Question 51 / 65 1 mark

S is a linearly independent subset of R^4. Select all true statements concerning S.

A. $|S| \leqslant 4$ ✓ **Correct**

B. Any subset of S is also a linearly independent subset of R^4 ✓ **Correct**

C. Any superset of S is also a linearly independent subset of R^4

D. S is a basis for R^4

Question 52 / 65 1 mark

Consider the matrix $\mathbf{M} = \begin{bmatrix} 2 & -1 \\ 3 & 1 \end{bmatrix}$. Which one of the following statements is true?

A. The eigenvalues of **M** are non-negative and real.

B. The eigenvalues of **M** are complex conjugate pairs. ✓ **Correct**

C. One eigenvalue of **M** is positive and real, and another eigenvalue of **M** is zero.

D. One eigenvalue of **M** is non-negative and real, and another eigenvalue of **M** is negative and real

Question 53 / 65 1 mark

Select all true statements.

A. The intersection of two subspaces of a vector space is a subspace. ✓ **Correct**

B. The union of two subspaces of a vector space is a subspace.

C. The empty set is a subspace of every vector space.

D. Every subset of a vector space need not be a subspace. ✓ **Correct**

Question 54 / 65 1 mark

Compute $\det(A)$ where $A = \begin{bmatrix} 1 & 2010 & 2020 \times 2030 \\ 1 & 2020 & 2030 \times 2010 \\ 1 & 2030 & 2010 \times 2020 \end{bmatrix}$

Answer: **2000**

Question 55 / 65 1 mark

Find the singular values of the following matrix:

$$\begin{bmatrix} -4 & -6 \\ 3 & -8 \end{bmatrix}$$

A. σ1 = 1, σ2 = 0

B. σ1 = 10, σ2 = 5 ✓ **Correct**

C. σ1 = 100, σ2 = 25

D. σ1 = σ2 = 0

Question 56 / 65 1 mark

If u, v, w are distinct vectors in R^2 that are pairwise orthogonal, which of the following are true?

A. u, v, w are linearly independent

B. u, v, w are linearly dependent ✓ **Correct**

C. Each of {u, v}, {v, w} and {u, w} is a basis for R^2

D. span{u, v, w} = R^3

Question 57 / 65 1 mark

Consider these two statements:

- S1: The sum of two symmetric matrices is a symmetric matrix.
- S2: The product of two symmetric matrices is a symmetric matrix.

A. S1 is true, S2 is false ✓ **Correct**

B. S1 is false, S2 is true

C. S1 and S2 are true

D. S1 and S2 are false

Question 58 / 65 1 mark

If $A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$, which of the following is A^8?

A. $\begin{bmatrix} 25 & 0 \\ 0 & 25 \end{bmatrix}$

B. $\begin{bmatrix} 125 & 0 \\ 0 & 125 \end{bmatrix}$

C. $\begin{bmatrix} 625 & 0 \\ 0 & 625 \end{bmatrix}$ ✓ **Correct**

D. $\begin{bmatrix} 3125 & 0 \\ 0 & 3125 \end{bmatrix}$

Question 59 / 65 1 mark

Consider a system of linear equations $PX = Q$ where $P \in \mathbb{R}^{3\times 3}$ and $Q \in \mathbb{R}^{3\times 1}$. Suppose P has an LU decomposition, $P = LU$, where

$$L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \quad \text{and} \quad U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}.$$

Which of the following statement(s) is/are true?

A. The system P X = Q can be solved by first solving LY = Q and then UX = Y . ✓ **Correct**

B. If P is invertible, then both L and U are invertible. ✓ **Correct**

C. If P is singular, then at least one of the diagonal elements of U is zero. ✓ **Correct**

D. If P is symmetric, then both L and U are symmetric.

Question 60 / 65 1 mark

Consider the given system of linear equations for some real k.

$x + ky = 1$
$kx + y = -1$

Which of the following is/are true?

A. There is exactly one value of k for which the above system of equations has no solution. ✓ **Correct**

B. There exist an infinite number of values of k for which the system of equations has no solution.

C. There exists exactly one value of k for which the system of equations has exactly one solution.

D. There exists exactly one value of k for which the system of equations has an infinite number of solutions. ✓ **Correct**

Question 61 / 65 1 mark

$\mathbf{w} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ is the weight vector of a linear classifier for a binary classification problem whose labels lie in the set $\{0, 1\}$. If the bias of the classifier is set to 0, which of the following is the equation of the decision boundary for this classifier?

A. $x_1 + 2x_2 + 3x_3 = 1$

B. $x_1 + 2x_2 + 3x_3 = -1$

C. $x_1 + 2x_2 + 3x_3 = 0$ ✓ **Correct**

D. $x_1 - 2x_3 + 3x_3 = 0$

Question 62 / 65 1 mark

Which of the following is the most appropriate description of the method used in PCA to achieve dimensionality reduction?

A. PCA achieves this by discarding a random subset of features in the dataset

B. PCA achieves this by selecting those features in the dataset along which the variance of the dataset is maximised

C. PCA achieves this by looking for those directions in the feature space along which the variance of the dataset is maximised ✓ **Correct**

D. PCA achieves this by retaining the those features in the dataset along which the variance of the dataset is minimised

Question 63 / 65 1 mark

Consider a covariance matrix $\mathbf{C}$ for a mean-centered dataset in $\mathbb{R}^3$. After performing standard PCA, the three principal components turn out to be $\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}$. Which of the following statements are true? You can assume that $\mathbf{C}$ is not the zero matrix.

A. $\mathbf{C}$ is a diagonal matrix. ✓ **Correct**

B. The diagonal entries of $\mathbf{C}$ are non-negative. ✓ **Correct**

C. The diagonal entries of $\mathbf{C}$ are strictly greater than zero.

D. $\mathbf{C}$ has to be the identity matrix.

Question 64 / 65 1 mark

Let Σ be the covariance matrix of a dataset, and suppose $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d$ are the eigenvalues of Σ. If we keep only the first k principal components, what is the fraction of total variance preserved by these k components?

A. $\frac{\lambda_1 + \lambda_2 + \cdots + \lambda_k}{\lambda_1 + \lambda_2 + \cdots + \lambda_d}$ ✓ **Correct**

B. $\frac{\lambda_1^2 + \lambda_2^2 + \cdots + \lambda_k^2}{\lambda_1^2 + \lambda_2^2 + \cdots + \lambda_d^2}$

C. $\frac{\lambda_k}{\lambda_1}$

D. $\frac{\lambda_k}{\lambda_1 + \lambda_2 + \cdots + \lambda_d}$

Question 65 / 65 1 mark

Which of the following is the loss for a single training example (**x**, y) for a linear regression model with weight vector **w**? Assume the most common scenario.

A. $\mathbf{w}^T x - y$

B. $(\mathbf{w}^T \mathbf{x} - y)^2$ ✓ **Correct**

C. $|(\mathbf{w}^T \mathbf{x} - y)|$

D. $-y \log(\mathbf{w}^T \mathbf{x})$